SQL CA 2021

Total possible marks : 180
qs : 22

Dr. A. Brennan

1. Be very careful when joining the employee and salary tables to get salary information/employee – join them on e.position = s.position and e.Branchno = s.Branchno
2. Just use **JOIN** for those Qs requiring joins

Tips

- Now()
- Datediff
- Group by … having …
- Limit x
- Order by a, b

1.

- Count the number of employees whose first names contain an a. Categorise them based on gender. Display the following;

Results
There are 8 first names with an a - grouped by gender = F
There are 11 first names with an a - grouped by gender = M

5 marks

2.

- Display the following using **now()**

CA
The SQL CA is taking place on Tuesday the 23rd of November

5 marks

3.

- Count the number of records where the first name starts with an **e** or the last name contains an **a**. Display the following:

Number of employees
9

5 marks

4.

- Display the following information from the salary table

Salary Info

The average salary for the 4 Manager employees is 87943.250000 whilst the minimum salary is 86323.00 and the maximum salary is 89456.00 compared to the overall salary average of 62086.166667

The average salary for the 4 HR employees is 54209.750000 whilst the minimum salary is 51048.00 and the maximum salary is 56783.00 compared to the overall salary average of 62086.166667

The average salary for the 4 Teller employees is 44105.500000 whilst the minimum salary is 41783.00 and the maximum salary is 46285.00 compared to the overall salary average of 62086.166667

- Note: you need to count the number of salary entries/category. After finding the average, max and min/category, compare them to the overall salary average regardless of category or position
- Look at the way it's ordered [only one order by]

10 marks

5.

Use CASE

- If the employee's bank is in Galway, it's 'The Hooker County'
- If the employee's bank is in Mayo, it's 'The Maritime County'
- If the employee's bank is in Tipperary, it's 'The Premier County'
- Else it's the 'Whatever you like yourself'

staffno	fname	lname	bank	county	County nicknames
1	Mary	White	AIB	Tipperary	The Premier County
2	John	Smith	BOI	Galway	The Hooker County
3	Eleanor	Aittin	AIB	Galway	The Hooker County
4	Tom	Smyth	AIB	Tipperary	The Premier County
5	Ellen	Bioxty	AIB	Tipperary	The Premier County
6	Sean	Reidin	AIB	Galway	The Hooker County
7	Liam	Skelton	AIB	Galway	The Hooker County
8	Mary	Duke	AIB	Tipperary	The Premier County
9	Sean	Fitzgerald	AIB	Galway	The Hooker County
10	Tim	Matthison	AIB	Tipperary	The Premier County
11	Sarah	Fluker	BOI	Mayo	The Maritime County
12	Ann	Davis	BOI	Mayo	The Maritime County
13	Kieran	Colby	BOI	Galway	The Hooker County
14	Liam	Alderson	AIB	Tipperary	The Premier County
15	Brendan	Goggins	BOI	Galway	The Hooker County
16	David	Kelly	AIB	Tipperary	The Premier County
17	Sarah	Jones	AIB	Tipperary	The Premier County
18	Lucy	Diamond	BOI	Galway	The Hooker County
19	Turlough	Icywater	BOI	Galway	The Hooker County
20	Harry	Potts	AIB	Galway	The Hooker County
21	Charlie	Smog	AIB	Tipperary	The Premier County
22	Andrew	Buxton	AIB	Galway	The Hooker County
23	Frances	Farmeril	BOI	Galway	The Hooker County
24	Gerald	Hoover	BOI	Galway	The Hooker County
25	Hattie	Smyth	BOI	Galway	The Hooker County

10 marks

6.

- Access every 3rd record from the employee table but only show the **TOP 6** records returned where the records are ordered by those employees most recently hired

Staffno	Fname	Lname	Gender	Position	Branchno	DateHired
3	Eleanor	Aittin	F	Manager	64789	2020-10-07
9	Sean	Fitzgerald	M	HR	64789	2020-08-17
6	Sean	Reidin	M	HR	64789	2020-07-08
21	Charlie	Smog	M	Manager	52786	2019-10-13
24	Gerald	Hoover	M	Teller	45556	2019-08-12
18	Lucy	Diamond	F	HR	45556	2019-08-06

5 marks

7.

- Show the sum of the salaries **by position** – watch the ordering at the end. Display the following:

Bank Position	Salary cost across the 4 branches
Manager	351773.00
HR	216839.00
Teller	176422.00

5 marks

8.

- Find the **top 5** most recent hires

Most recent hires
David Kelly was hired in the last 106 days
Tom Smyth was hired in the last 133 days
Eleanor Aittin was hired in the last 412 days
Ellen Bioxty was hired in the last 426 days
Sean Fitzgerald was hired in the last 463 days

10 marks

9.

- Find the difference in days from the date hired to today for all females categorised by position. Watch the ordering

position	Recent Female hires by Position
Manager	3684
HR	2463
Teller	2337

10 marks

10.

- Count the number of employees by position, gender and branchno for all of those employees who are either working in branch no **52786** or branch no **64789**. Display the following:

position	gender	branchno	Number of Employees
HR	F	52786	1
HR	M	52786	1
HR	M	64789	2
Manager	F	52786	1
Manager	F	64789	1
Manager	M	52786	2
Manager	M	64789	1
Teller	F	52786	2
Teller	M	52786	2
Teller	M	64789	2

10 marks

11.

- Display the following information. Note: you will be joining 3 tables. When you join the employee and salary table, join them on e.position = s.position and e.Branchno = s.Branchno. Watch the way the output is ordered

staffno	fname	lname	position	branchno	bank	county	salary
1	Mary	White	Teller	52786	AIB	Tipperary	45367.00
2	John	Smith	Manager	45556	BOI	Galway	88543.00
3	Eleanor	Aittin	Manager	64789	AIB	Galway	86323.00
4	Tom	Smyth	Teller	52786	AIB	Tipperary	45367.00
5	Ellen	Bioxty	HR	52786	AIB	Tipperary	53765.00
6	Sean	Reidin	HR	64789	AIB	Galway	55243.00
7	Liam	Skelton	Manager	64789	AIB	Galway	86323.00
8	Mary	Duke	Teller	52786	AIB	Tipperary	45367.00
9	Sean	Fitzgerald	HR	64789	AIB	Galway	55243.00
10	Tim	Matthison	Teller	52786	AIB	Tipperary	45367.00
11	Sarah	Fluker	Manager	87451	BOI	Mayo	87451.00
12	Ann	Davis	HR	87451	BOI	Mayo	51048.00
13	Kieran	Colby	Teller	45556	BOI	Galway	41783.00
14	Liam	Alderson	Manager	52786	AIB	Tipperary	89456.00
15	Brendan	Goggins	HR	45556	BOI	Galway	56783.00
16	David	Kelly	HR	52786	AIB	Tipperary	53765.00
17	Sarah	Jones	Manager	52786	AIB	Tipperary	89456.00
18	Lucy	Diamond	HR	45556	BOI	Galway	56783.00
19	Turlough	Icywater	Manager	45556	BOI	Galway	88543.00
20	Harry	Potts	Teller	64789	AIB	Galway	42987.00
21	Charlie	Smog	Manager	52786	AIB	Tipperary	89456.00
22	Andrew	Buxton	Teller	64789	AIB	Galway	42987.00
23	Frances	Farmeril	Manager	45556	BOI	Galway	88543.00
24	Gerald	Hoover	Teller	45556	BOI	Galway	41783.00
25	Hattie	Smyth	Teller	45556	BOI	Galway	41783.00

10 marks

12.

- Add a concat to the previous SQL query (#11) to display:

Employee Info
Mary White works as a Teller in AIB:Tipperary at a salary of : 45367.00
John Smith works as a Manager in BOI:Galway at a salary of : 88543.00
Eleanor Aittin works as a Manager in AIB:Galway at a salary of : 86323.00
Tom Smyth works as a Teller in AIB:Tipperary at a salary of : 45367.00
Ellen Bioxty works as a HR in AIB:Tipperary at a salary of : 53765.00
Sean Reidin works as a HR in AIB:Galway at a salary of : 55243.00
Liam Skelton works as a Manager in AIB:Galway at a salary of : 86323.00
Mary Duke works as a Teller in AIB:Tipperary at a salary of : 45367.00
Sean Fitzgerald works as a HR in AIB:Galway at a salary of : 55243.00
Tim Matthison works as a Teller in AIB:Tipperary at a salary of : 45367.00
Sarah Fluker works as a Manager in BOI:Mayo at a salary of : 87451.00
Ann Davis works as a HR in BOI:Mayo at a salary of : 51048.00
Kieran Colby works as a Teller in BOI:Galway at a salary of : 41783.00
Liam Alderson works as a Manager in AIB:Tipperary at a salary of : 89456.00
Brendan Goggins works as a HR in BOI:Galway at a salary of : 56783.00
David Kelly works as a HR in AIB:Tipperary at a salary of : 53765.00
Sarah Jones works as a Manager in AIB:Tipperary at a salary of : 89456.00
Lucy Diamond works as a HR in BOI:Galway at a salary of : 56783.00
Turlough Icywater works as a Manager in BOI:Galway at a salary of : 88543.00
Harry Potts works as a Teller in AIB:Galway at a salary of : 42987.00
Charlie Smog works as a Manager in AIB:Tipperary at a salary of : 89456.00
Andrew Buxton works as a Teller in AIB:Galway at a salary of : 42987.00
Frances Farmeril works as a Manager in BOI:Galway at a salary of : 88543.00
Gerald Hoover works as a Teller in BOI:Galway at a salary of : 41783.00
Hattie Smyth works as a Teller in BOI:Galway at a salary of : 41783.00

8 marks

13.

Revise SQL query #11 ;

- Add an IF so that if the person's salary is less than the average salary for that position (regardless of branchno), then 'Less than the average salary for this position' is displayed
- Else 'More than the average salary for this position' is displayed

staffno	fname	lname	position	branchno	bank	county	salary	SALARY INFO
1	Mary	White	Teller	52786	AIB	Tipperary	45367.00	More than the average salary for this position
2	John	Smith	Manager	45556	BOI	Galway	88543.00	More than the average salary for this position
3	Eleanor	Aittin	Manager	64789	AIB	Galway	86323.00	Less than the average salary for this position
4	Tom	Smyth	Teller	52786	AIB	Tipperary	45367.00	More than the average salary for this position
5	Ellen	Bioxty	HR	52786	AIB	Tipperary	53765.00	Less than the average salary for this position
6	Sean	Reidin	HR	64789	AIB	Galway	55243.00	More than the average salary for this position
7	Liam	Skelton	Manager	64789	AIB	Galway	86323.00	Less than the average salary for this position
8	Mary	Duke	Teller	52786	AIB	Tipperary	45367.00	More than the average salary for this position
9	Sean	Fitzgerald	HR	64789	AIB	Galway	55243.00	More than the average salary for this position
10	Tim	Matthison	Teller	52786	AIB	Tipperary	45367.00	More than the average salary for this position
11	Sarah	Fluker	Manager	87451	BOI	Mayo	87451.00	Less than the average salary for this position
12	Ann	Davis	HR	87451	BOI	Mayo	51048.00	Less than the average salary for this position
13	Kieran	Colby	Teller	45556	BOI	Galway	41783.00	Less than the average salary for this position
14	Liam	Alderson	Manager	52786	AIB	Tipperary	89456.00	More than the average salary for this position
15	Brendan	Goggins	HR	45556	BOI	Galway	56783.00	More than the average salary for this position
16	David	Kelly	HR	52786	AIB	Tipperary	53765.00	Less than the average salary for this position
17	Sarah	Jones	Manager	52786	AIB	Tipperary	89456.00	More than the average salary for this position
18	Lucy	Diamond	HR	45556	BOI	Galway	56783.00	More than the average salary for this position
19	Turlough	Icywater	Manager	45556	BOI	Galway	88543.00	More than the average salary for this position
20	Harry	Potts	Teller	64789	AIB	Galway	42987.00	Less than the average salary for this position
21	Charlie	Smog	Manager	52786	AIB	Tipperary	89456.00	More than the average salary for this position
22	Andrew	Buxton	Teller	64789	AIB	Galway	42987.00	Less than the average salary for this position
23	Frances	Farmeril	Manager	45556	BOI	Galway	88543.00	More than the average salary for this position
24	Gerald	Hoover	Teller	45556	BOI	Galway	41783.00	Less than the average salary for this position
25	Hattie	Smyth	Teller	45556	BOI	Galway	41783.00	Less than the average salary for this position

10 marks

14.

- Add in a new column to SQL query #13 , called 'avg salary for this position' which checks for the average salary for this position regardless of branchno

staffno	fname	lname	position	salary	avg salary for this position	SALARY INFO
1	Mary	White	Teller	45367.00	44105.500000	More than the average salary for this position
2	John	Smith	Manager	88543.00	87943.250000	More than the average salary for this position
3	Eleanor	Aittin	Manager	86323.00	87943.250000	Less than the average salary for this position
4	Tom	Smyth	Teller	45367.00	44105.500000	More than the average salary for this position
5	Ellen	Bioxty	HR	53765.00	54209.750000	Less than the average salary for this position
6	Sean	Reidin	HR	55243.00	54209.750000	More than the average salary for this position
7	Liam	Skelton	Manager	86323.00	87943.250000	Less than the average salary for this position
8	Mary	Duke	Teller	45367.00	44105.500000	More than the average salary for this position
9	Sean	Fitzgerald	HR	55243.00	54209.750000	More than the average salary for this position
10	Tim	Matthison	Teller	45367.00	44105.500000	More than the average salary for this position
11	Sarah	Fluker	Manager	87451.00	87943.250000	Less than the average salary for this position
12	Ann	Davis	HR	51048.00	54209.750000	Less than the average salary for this position
13	Kieran	Colby	Teller	41783.00	44105.500000	Less than the average salary for this position
14	Liam	Alderson	Manager	89456.00	87943.250000	More than the average salary for this position
15	Brendan	Goggins	HR	56783.00	54209.750000	More than the average salary for this position
16	David	Kelly	HR	53765.00	54209.750000	Less than the average salary for this position
17	Sarah	Jones	Manager	89456.00	87943.250000	More than the average salary for this position
18	Lucy	Diamond	HR	56783.00	54209.750000	More than the average salary for this position
19	Turlough	Icywater	Manager	88543.00	87943.250000	More than the average salary for this position
20	Harry	Potts	Teller	42987.00	44105.500000	Less than the average salary for this position
21	Charlie	Smog	Manager	89456.00	87943.250000	More than the average salary for this position
22	Andrew	Buxton	Teller	42987.00	44105.500000	Less than the average salary for this position
23	Frances	Farmeril	Manager	88543.00	87943.250000	More than the average salary for this position
24	Gerald	Hoover	Teller	41783.00	44105.500000	Less than the average salary for this position
25	Hattie	Smyth	Teller	41783.00	44105.500000	Less than the average salary for this position

8 marks

15.

- Display the following information for all employees whose salary is greater than **ALL** of the average salaries categorised by branchno. Watch the ordering

staffno	fname	lname	salary
14	Liam	Alderson	89456.00
17	Sarah	Jones	89456.00
21	Charlie	Smog	89456.00
23	Frances	Farmeril	88543.00
19	Turlough	Icywater	88543.00
2	John	Smith	88543.00
11	Sarah	Fluker	87451.00
3	Eleanor	Aittin	86323.00
7	Liam	Skelton	86323.00

10 marks

16.

- Display the following information for all employees who were hired more recently than 2020-01-01 (use >). If they are a Manager, they get a bonus of 5%. If they are a Teller, they get a 30% bonus otherwise, as a HR employee, they get a 25% bonus (use CASE)

Name	Surname	Bank Position	Bonus Scheme	salary	New Salary + Bonus
Mary	White	Teller	Teller gets a 30% bonus	45367.00	58977.10
John	Smith	Manager	Manager gets a 5% bonus	88543.00	92970.15
Eleanor	Aittin	Manager	Manager gets a 5% bonus	86323.00	90639.15
Tom	Smyth	Teller	Teller gets a 30% bonus	45367.00	58977.10
Ellen	Bioxty	HR	HR gets a 25% bonus	53765.00	67206.25
Sean	Reidin	HR	HR gets a 25% bonus	55243.00	69053.75
Sean	Fitzgerald	HR	HR gets a 25% bonus	55243.00	69053.75
Sarah	Fluker	Manager	Manager gets a 5% bonus	87451.00	91823.55
David	Kelly	HR	HR gets a 25% bonus	53765.00	67206.25
Sarah	Jones	Manager	Manager gets a 5% bonus	89456.00	93928.80
Andrew	Buxton	Teller	Teller gets a 30% bonus	42987.00	55883.10

13 marks

17.

- Show employee and branch information as follows;
- Look at how the data is presented

fname	lname	position	branchno	bank	county
Eleanor	Aittin	Manager	64789	AIB	Galway
Andrew	Buxton	Teller	64789	AIB	Galway
Sean	Fitzgerald	HR	64789	AIB	Galway
Harry	Potts	Teller	64789	AIB	Galway
Sean	Reidin	HR	64789	AIB	Galway
Liam	Skelton	Manager	64789	AIB	Galway
Liam	Alderson	Manager	52786	AIB	Tipperary
Ellen	Bioxty	HR	52786	AIB	Tipperary
Mary	Duke	Teller	52786	AIB	Tipperary
Sarah	Jones	Manager	52786	AIB	Tipperary
David	Kelly	HR	52786	AIB	Tipperary
Tim	Matthison	Teller	52786	AIB	Tipperary
Charlie	Smog	Manager	52786	AIB	Tipperary
Tom	Smyth	Teller	52786	AIB	Tipperary
Mary	White	Teller	52786	AIB	Tipperary
Kieran	Colby	Teller	45556	BOI	Galway
Lucy	Diamond	HR	45556	BOI	Galway
Frances	Farmeril	Manager	45556	BOI	Galway
Brendan	Goggins	HR	45556	BOI	Galway
Gerald	Hoover	Teller	45556	BOI	Galway
Turlough	Icywater	Manager	45556	BOI	Galway
John	Smith	Manager	45556	BOI	Galway
Hattie	Smyth	Teller	45556	BOI	Galway
Ann	Davis	HR	87451	BOI	Mayo
Sarah	Fluker	Manager	87451	BOI	Mayo

Tip : join + multiple orders

8 marks

18.

- Show the following employee and salary information for those employees whose salary is greater than **ANY** of the average salaries categorised by branch no AND less than **ANY** of the minimum salaries categorised by position

fname	lname	position	position	branchno	salary
John	Smith	Manager	Manager	45556	88543.00
Turlough	Icywater	Manager	Manager	45556	88543.00
Frances	Farmeril	Manager	Manager	45556	88543.00
Liam	Alderson	Manager	Manager	52786	89456.00
Sarah	Jones	Manager	Manager	52786	89456.00
Charlie	Smog	Manager	Manager	52786	89456.00
Eleanor	Aittin	Manager	Manager	64789	86323.00
Liam	Skelton	Manager	Manager	64789	86323.00
Sarah	Fluker	Manager	Manager	87451	87451.00

Tip : use subqueries

10 marks

19.

- Show the following employee and bank information

Employee Details
Mary White works at AIB, Clonmel, Tipperary
John Smith works at BOI, Claregalway, Galway
Eleanor Aittin works at AIB, Oranmore, Galway
Tom Smyth works at AIB, Clonmel, Tipperary
Ellen Bioxty works at AIB, Clonmel, Tipperary
Sean Reidin works at AIB, Oranmore, Galway
Liam Skelton works at AIB, Oranmore, Galway
Mary Duke works at AIB, Clonmel, Tipperary
Sean Fitzgerald works at AIB, Oranmore, Galway
Tim Matthison works at AIB, Clonmel, Tipperary
Sarah Fluker works at BOI, Castlebar, Mayo
Ann Davis works at BOI, Castlebar, Mayo
Kieran Colby works at BOI, Claregalway, Galway
Liam Alderson works at AIB, Clonmel, Tipperary
Brendan Goggins works at BOI, Claregalway, Galway
David Kelly works at AIB, Clonmel, Tipperary
Sarah Jones works at AIB, Clonmel, Tipperary
Lucy Diamond works at BOI, Claregalway, Galway
Turlough Icywater works at BOI, Claregalway, Galway
Harry Potts works at AIB, Oranmore, Galway
Charlie Smog works at AIB, Clonmel, Tipperary
Andrew Buxton works at AIB, Oranmore, Galway
Frances Farmeril works at BOI, Claregalway, Galway
Gerald Hoover works at BOI, Claregalway, Galway
Hattie Smyth works at BOI, Claregalway, Galway

Tip : 2 joins

5 marks

20.

- Count the number of employees per position but only show those positions who have more than 7 employees. Watch the ordering

position 1	Number of employees in this position
Teller	9
Manager	9

8 marks

21.

- Display the following for all employees whose salary * 1.5 is greater than **ALL** of the list of results returned when looking for the average salary grouped by position. Watch the ordering

Name	Surname	Position	Branchno	Salary
Eleanor	Aittin	Manager	64789	86323.00
Liam	Alderson	Manager	52786	89456.00
Frances	Farmeril	Manager	45556	88543.00
Sarah	Fluker	Manager	87451	87451.00
Turlough	Icywater	Manager	45556	88543.00
Sarah	Jones	Manager	52786	89456.00
Liam	Skelton	Manager	64789	86323.00
John	Smith	Manager	45556	88543.00
Charlie	Smog	Manager	52786	89456.00

10 marks

22.

- Show the following information for all of those employees who are not in banks with a branch code of 52786 and 87451. Use **NOT** and **IN** (watch the ordering)

lname	fname	bank	county
Smyth	Hattie	BOI	Galway
Smith	John	BOI	Galway
Skelton	Liam	AIB	Galway
Reidin	Sean	AIB	Galway
Potts	Harry	AIB	Galway
Icywater	Turlough	BOI	Galway
Hoover	Gerald	BOI	Galway
Goggins	Brendan	BOI	Galway
Fitzgerald	Sean	AIB	Galway
Farmeril	Frances	BOI	Galway
Diamond	Lucy	BOI	Galway
Colby	Kieran	BOI	Galway
Buxton	Andrew	AIB	Galway
Aittin	Eleanor	AIB	Galway

5 marks